

FORUM & POLICY

Collecting insects to conserve them: a call for ethical caution

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Abstract. 1. Insect sampling for the purpose of measuring biodiversity – as well as entomological research more generally – largely assumes that insects lack consciousness. Here, we briefly present some arguments that insects are conscious and encourage entomologists to revisit their ethical codes in light of them.

2. Specifically, we adapt the Three Rs, guidelines proposed in 1959 by WMS Russell and RL Burch that have become the dominant way of thinking about the ethics of using animals in research.

3. The Three Rs specify the need to replace, reduce, and refine the use of animals in research, yet have received little attention in entomological circles, which is perhaps unsurprising given that Russell and Burch explicitly excluded invertebrates from their purview.

4. As a specific case, we consider issues of suffering and bycatch in the use of Malaise traps for insect sampling.

5. While we do not claim that entomologists have an obligation to adopt the Three Rs framework wholesale, we do suggest that there is reason to adopt it in a modified form to mitigate moral risk especially in the context of conservation.

Key words. 3Rs, consciousness, ethics, insect collection.

Introduction

Insect conservation is driven by many considerations: the value of insects in their own right, their environmental value, and the many benefits that they provide to human beings, ranging from the aesthetic to the highly practical (e.g., medical research). However, even when human beings value insects, the focus tends to be solely on the species rather than on individuals. By contrast, consider the case of vertebrate organisms: we value individual giraffes as instances of a particular kind, but we also value them as individuals, so that harming a particular individual requires some justification. (We say much more about this below.) With this forum paper, we wish to question this tendency in the case of insects. In particular, we consider whether recent insights into insect consciousness provide entomologists with reasons to adjust ethical codes, and so

their practices. For the sake of space, we focus on the ethics of collecting insects, though we briefly discuss the implications for entomological research more generally [noting that Freelance (2018) considers the case of animal experimentation in depth]. In particular, we consider here whether the killing of so many organisms can be justified, especially given that insect sampling often kills large numbers of individuals indiscriminately. In addition to ‘selective’ sampling using insect nets, for example, entomologists sample insects using Malaise nets (which collect, then poison, flying insects) and pitfall traps (which collect, then drown, walking insects). In other instances, entomologists have applied insecticides to entire tropical trees to provide an estimate of insect species diversity in tropical rainforest (e.g., Paarmann and Stork 1987).

Our main aim here is to argue that entomologists have reason to adopt a modified version of the 3Rs guidelines, first proposed by Russell and Burch (1959). According to the 3Rs, those involved in research with animals should strive to (i) *replace* animals with non-living models, (ii) *reduce* the number of animals used in research, and (iii)

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refine animal care and use practices so that animals are better off. Obviously, there will be many contexts in which animals cannot be replaced by, say, computer models, or where refinement is impossible given the specific research protocol. The claim is not that these changes should always be made, but rather that the researcher should always be looking for ways to reduce the negative impact of their research on their subjects (And clearly, an assessment of whether an animal is 'better off' should be based on the best available evidence – rather than, for example, anthropomorphism.).

The 3Rs are regularly applied to vertebrates (for a standard presentation for those involved in animal experimentation, see <https://www.nc3rs.org.uk/the-3rs>), but they have not been applied to insects. This is not surprising, not least because Russell and Burch (1959) explicitly excluded invertebrates from the scope of their study. They wrote:

...we have restricted the discussion entirely to the vertebrates... The higher invertebrates perhaps deserve a review to themselves, but they raise many problems which would gravely complicate an account which can otherwise be quite general and confident. Only one group, the insects, are of any numerical consequence in practice [... because many] experiments on insects are concerned with the development of substances that can be used to kill them ... The privileged status of vertebrates may appear arbitrary when [we] compare, say, lamprey with octopus; but for simplicity and clarity, we shall stick to the traditional division, which has much to recommend it.

As the rest of their book makes clear, Russell and Burch (1959) based this restriction on the thought that most invertebrates are not conscious, and so cannot experience pain or otherwise feel distress.

That thought was plausible in 1959, and it remains plausible now. However, there is recent research that should give us pause. There is insufficient evidence for the claim that insects are conscious (i.e., have sentience, subjective experiences, or the capacity for experiential states), yet there are now some interesting arguments for that claim, and the possibility that insects are conscious raises challenging ethical questions. Here, in the spirit of an 'ecological ethics' (Minteer & Collins, 2008), we present those arguments and encourage entomologists to revisit their ethical codes in light of them.

Preliminaries

Before saying anything at all about the evidence pertaining to insects being conscious, we begin by clarifying the issue at stake.

First, why focus on consciousness? Part of the answer should be clear from the above: consciousness matters because non-conscious beings cannot experience pain, and

pain is morally significant. Additionally, though, the presence or absence of consciousness seems to be the difference between whether a being has, or does not have, morally relevant interests. In some sense, of course, everything has interests: a houseplant needs sunlight to grow, and so we can say that the houseplant has an interest in getting sunlight. However, it is also true that for salt crystals to grow, they need to be undisturbed. So, we might say that salt crystals have an interest in inhabiting vibration-free environments. Once we recognize this, we realize that we need to draw a distinction between those interests that deserve moral consideration all on their own — quite independently of whether any person might have some economic or aesthetic or other concern for the satisfaction of those interests — and those that do not. In the case of the salt crystals, it seems plausible that we only have a moral reason to provide them with vibration-free environments if someone cares about growing salt crystals. If no one does, then we can happily create all the vibrations we want. Why is this? A natural thought is that it is because the salt crystals are not conscious: they do not care about whether they grow.

As the above suggests, there are at least two arguments for this idea. One of them is based on our considered judgments about cases. If we do not say that consciousness is the line between having morally relevant interests and not having them, then it will be very difficult to deny that inanimate objects have morally relevant interests, since there is some sense in which things can go better and worse for them. This is the point of the salt example, but it is easy to come up with others. Consider, for instance, the question of whether we distinctly have a *moral* reason to put oil in an engine — which, we might say, has an interest in having adequate oil — as opposed to a merely prudential or practical reason to do it. Another argument is based on a theory about the point of morality. It seems plausible that morality is about giving due consideration to other beings. However, we might want an explanation for this. Why, exactly, does any being deserve consideration at all? In some cases, this question is very difficult to answer. For instance, if no one cares about a houseplant — if no person or animal has any economic or aesthetic or nutritional interest in the plant — then what difference does it make whether it is given adequate sunlight? After all, the plant itself does not care how it is treated. This makes it difficult to explain why we owe it any consideration. By contrast, when we consider human beings and many non-human animals, they clearly do care how they are treated — a capacity they have in virtue of being conscious. So, the presence or absence of consciousness seems to determine whether a being's interests are morally relevant.

Obviously, there is far more to say about these issues — see Gruen (2017) for further discussion — but we will take this point for granted in what follows. This is because even if the philosophical point is mistaken, it remains that the presence or absence of consciousness is the factor that explains when research is subject to a

heightened degree of moral scrutiny. Even if non-conscious beings have morally relevant interests, the interests of conscious beings appear to be especially important, as reflected in the ethics codes that guide research involving non-human animals. And when we consider the ethics of such research, we should notice that it is complicated by two factors. First, there is uncertainty about the specific mental capacities of non-human animals. For instance, while there is no debate about whether mice can feel pain, there is debate about the degree to which mice are self-aware. This uncertainty makes it difficult to assess how, exactly, mice can be harmed. Do they have desires about their own futures that humans can frustrate? Or are their psychologies far simpler, such that they can only be harmed by the experience of pain? Second, there is uncertainty about how bad it is, morally speaking, to harm non-human animals. This means that while, for instance, there is general agreement that it is morally bad to cause pain in non-human animals, there is considerable debate about how easy is to justify causing pain. Do significant human goods need to be on the line – such as a major breakthrough in cancer research? Or is the mere pursuit of knowledge sufficiently important to justify any harm that research might inflict on its non-human subjects?

One response to this uncertainty is to doubt the possibility of developing a reasonable ethic for research involving non-human animals. But such doubt is both unjustified and professionally irresponsible. It is unjustified because researchers do not need to have all the answers in order to agree on a limited set of guidelines. It is professionally irresponsible because researchers have an obligation to confront the practical and theoretical questions that their research raises. The history of science includes all too many cautionary tales, so society now quite rightly insists that scientists attend to the moral complexities of their work.

However, even if it seems possible to develop a reasonable ethic for research involving non-human animals, there remain difficult questions about how to proceed. We begin by considering two fundamental issues: first, baseline expectations about which there is little reasonable disagreement; second, strategies for mitigating moral risk where there is considerable reasonable disagreement. Let us explain.

The baseline expectations are fairly straightforward. These are the sorts of norms that follow from agreed-upon values. Consider that there is now a broad consensus to the effect that species conservation is an important goal. It follows that research should not employ techniques that threaten entire species. If, for instance, a species of beetle has only been found in one cave system, it would be irresponsible to collect every beetle from that system, since that would lead to extinction of the species.

Mitigating moral risk is a more complex endeavour. It involves considering the various ways in which we might be wrong about the empirical facts and about what matters morally. On the empirical side, we might think that the members of a certain species have only a very limited

ability to engage in mental time travel – namely, the ability to imagine past or future state of affairs. However, we could be mistaken about that, and so be mistaken about the ways in which those individuals can be harmed. Beings that can imagine themselves in the future can have more complex desires about their futures, and so can have those desires frustrated in ways that can be bad for them. As a concrete example, if a person can imagine being able to see his daughter graduate from college, then he can want to experience that event in reality, and so it can be bad for him if circumstances prevent him from experiencing it. On the moral side, we might think that the basic interests of human beings always outweigh the basic interests of non-human animals, and yet we could be mistaken about that, and therefore be mistaken about when it is permissible to use non-human animals in research. Of course, it is often difficult – if not impossible – to assign specific probabilities to such possibilities, and the imprecision creates a lot of room for debate about the appropriate degree of caution. Still, there are familiar ways in which individual researchers engage in risk mitigation. For example, many entomologists freeze the insects that they plan to study, not necessarily because they are convinced that insects *can* feel pain, but because they think that it is *possible*. And *if* insects can feel pain do, then freezing them is thought to be a better way to kill them than some of the alternatives. These sorts of practices can provide helpful models for thinking about how moral risks can be managed.

Philosophers have long argued that humans undervalue the lives and experiences of non-human animals (see, for instance, Singer, 1975; Regan, 1983; Taylor 1986, and DeGrazia, 1996). So, even if there is only very weak evidence in favour of attributing consciousness to insects, we may have reason to be much more cautious in our treatment of insects simply because humans tend to give animals – and insects, especially – far too little weight in moral deliberations. However, we will not explore this issue here. Instead, we will simply focus on the empirical question, arguing that the evidence is sufficient to justify a reconsideration of ethical codes for the case of insect collection.

Are insects conscious?

We have already indicated that it is reasonable to deny that insects are conscious. Our aim is just to show that insect consciousness is a live possibility (see also Lockwood, 1987; Freelance, 2018). That point enough to be able to argue that there should be some adjustments to the ethical codes that entomologists endorse.

Insects demonstrate surprisingly complex behaviour, much of which is now well-known. As examples, earwigs display maternal care (Köllicker and Vancassel 2007), honeybees can find artificial food sources that have been moved significant distances by experimenters (Griffin, 1984), and weaver ants can recruit other ants to fight

without themselves having been exposed to the threat (Hölldobler & Wilson, 1978). Elsewhere, Bateson *et al.* (2011) have argued that bees can come to interpret ambiguous stimuli negatively, a phenomenon caused by negative emotions in humans. However, there remains a tendency to apply a particularly stringent version of Morgan's Canon, according to which we should deny that insects are conscious as long as it is possible to explain their behaviour without postulating conscious states. There are serious problems with this idea – e.g., it ignores the fact that the explanation in terms of conscious states may be the better one (Fitzpatrick, 2018) – but since it remains the default approach, more pointed evidence is required.

Presumably, if insects can feel pain, then they are conscious. But can they feel pain? The evidence is difficult to interpret. Insects have neurons that respond to negative stimuli, and some insects react to analgesics in ways that suggest these negative signals can be blocked. Zabala and Gómez (1991), for instance, found that cockroaches left a heated box more slowly after being injected with morphine, and Balderrama *et al.* (1987) showed that receiving morphine caused bees to sting less aggressively after having a round of electrical shocks. What's more, it does not look like morphine is simply decreasing behavioural responsiveness generally. For instance, fruit flies tend to move towards light, but if they are placed in a glass tube and the centre of that tube is heated, they tend not to pass through the hot portion. However, if administered an analgesic, they are willing to pass through that otherwise-uncomfortable spot, again reaching the light source (Dimitrijevic *et al.*, 2005). Fruit flies also act in ways that suggest that they can learn to avoid painful stimuli: they can come to associate electric shock with an odour, and then will avoid that odour for up to 24 h afterward (Yarali *et al.*, 2008).

Nevertheless, it is also the case that insects often fail to respond to damage in the way that humans might expect were they to be suffering. In the words of an influential review by Eisemann *et al.* (1984: 166):

...our experience has been that insects will continue with normal activities even after severe injury or removal of body parts. An insect walking with a crushed tarsus, for example, will continue applying it to the substrate with undiminished force. Among our other observations are those on a locust which continued to feed whilst itself being eaten by a mantis; aphids continuing to feed whilst being eaten by coccinellids; a tsetse fly which flew in to feed although half-dissected; caterpillars which continue to feed whilst tachinid larvae bore into them; many insects which go about their normal life whilst being eaten by large internal parasitoids; and male mantids which continue to mate as they are eaten by their partners. Insects show no immobilisation equivalent to the mammalian reaction to painful body damage, nor have our preliminary observations of the response of

locusts to bee stings revealed anything analogous to a mammalian response.

Of course, many animals have systems that block pain – in humans, as well as some insects, endorphins play that role (Duve *et al.*, 1979; Remy *et al.*, 1979) – and it is possible that insects have a similar system that allows them to maintain relatively normal function despite severe damage. Moreover, in circumstances where the odds of survival are sufficiently low, it may be in the best interest of an insect to simply stay the course: this might allow mating to take place when the opportunity is available and the continuance of other behaviours (e.g., feeding) might simply be a byproduct of that fitness-enhancing mechanism. Furthermore, chitin does not regrow, which means that insects do not have the same incentive to protect damaged part of their bodies that, say, mammals do: a dog's leg can heal, so it makes sense to try to save a damaged limb; however, an ant's leg will not heal, so it may as well be abandoned.

Viewed in isolation, these competing considerations are difficult to resolve. But there are also broader theoretical reasons to suggest that insects are conscious, which make it more plausible that the best interpretation of pain-suggesting behaviours are, in fact, experiences of pain. Consider, for instance, the work of Klein and Barron (2016), who draw on Merker's (2005) theory of consciousness. Merker (2005) argues that consciousness arises from the need to integrate information from a variety of sources, and that this happens when organisms become mobile and have to represent their location in space. Klein and Barron (2016: 7–8) point out that:

...there are significant parallels between the functional organisation of the insect brain and that of the vertebrate midbrain behavioural core control system. Both systems have specialised regions for processing the position of the moving animal in space. In both systems action selection is resolved by combining information on position with information on the environment, the relevance of stimuli in the environment to the animal, and the state of the animal. In vertebrates the channels of information are sufficiently unified such that the system as a whole creates a functional representation of the state of the mobile animal in space as a solution for effective decision-making.

As we have argued, processing of this kind supports the capacity for a subjective experience of the environment. Processing in the insect brain is unified to a similar degree, for similar reasons. Hence, we propose that the insect brain can also support a capacity for subjective experience.

In other words, we find systems that seem to fulfil the same function in vertebrates and invertebrates, and in the

case of vertebrates, it is plausible that the relevant system explains the presence of conscious awareness in the organism. This simulation is highly valuable to mobile organisms that find themselves having to integrate various sources of information – visual, auditory, etc. – so that they can navigate their environment. And, if it fulfils the same function in invertebrates, that is a reason to think that they have some degree of consciousness as well. (For further discussion of this conclusion, see the special issue of *Animal Sentience* devoted to Klein and Barron's essay.)

Merker's theory is hardly the only one in the field. Some insist that consciousness requires higher-order thought (Edelman, 2003); others maintain that it requires awareness of one's self as a subject (Morin, 2006). But it is certainly a prominent theory, and Merker is one of many who argue that certain integrated structures in the vertebrate midbrain support conscious experience (e.g., Damasio & Carvalho, 2013; Mashour & Alkire, 2013). Moreover, Klein and Barron (2016) provide good reasons for applying this approach to insects. Does this establish that insects are, in fact, conscious? Again, no. Yet this line of reasoning raises the probability that insects are conscious, and that makes it appropriate to give that possibility some weight in our moral deliberations. We next show, however, that existing entomological ethical codes have generally ignored it.

Survey of guidelines for insect sampling

Plainly, entomologists do give insects some weight in their moral deliberations. However, existing ethical codes generally value insects either instrumentally or as exemplars of the species. The codes do not indicate that individual insects deserve moral consideration.

Consider, for instance, a case that is particularly pertinent to the conservation of insect biodiversity, the ethics of sampling insects. We readily acknowledge that there are many good reasons for entomologists to collect insects. Collection is necessary to document the planet's biodiversity, which is one basis for determining how to protect it. As with plant herbaria, long-term insect collections also allow for the long-term monitoring of environmental change – and such voucher records have often provided insights that were unforeseen.

Nonetheless, insect sampling ethics tend to be driven by anthropocentric concerns (e.g., do not sample in ways that would hinder colleagues' research), environmental responsibility (e.g., do not upset the ecosystem any more than necessary), and species-level considerations (e.g., do not kill so many members of a particular population that its existence is threatened). Collecting codes designed to rein in indiscriminate sampling share features such as the following: limit the number of specimens collected (e.g., minimise year-to-year collections at the same site); minimise or avoid collection of rare species; examine live specimens whenever possible (e.g., large, easily identifiable species); consider photographic documentation, depending

on purpose; minimise damage to the habitat generally; collect adequate data; and safeguard long-term storage and maintenance of specimens (e.g., Joint Committee for the Conservation of British Invertebrates 2002; Buglife 2013; ESA 2016; TEA n/d). However, they omit concern about insect lives.

Consider two additional examples. In his encyclopedic photographic guide to insects of eastern North America, Marshall (2006: 613) describes how to make an insect collection and then provides 'a few reasons not to make an insect collection.' He focuses on practical considerations related to the time and expense required to make a proper insect collection, as well as its illegality in many places, having summarily discounted ecological and ethical arguments. With regard to the ecological argument, he states that 'the killing of a few hundred (or even a few hundred thousand) adult insects has absolutely no impact on insect populations.' [This statement is clearly problematic, not least because some species are rare, as recognised by codes such as the Joint Committee for the Conservation of British Invertebrates (2002), which states that 'even an apparently common species may be locally vulnerable.'] Marshall then discounts ethical concerns with the claim that 'it really does not make much sense to be squeamish about killing a few specimens in a killing jar or freezer when you directly kill thousands on your car radiator and indirectly kill millions by buying cotton or almost anything in a grocery store' – a clear expression of how individual insect lives are discounted that also obviates the distinction between intentional and unintentional killing.

Elsewhere, Wilson and Carril (2016: 36), focusing on bees, defend insect collection by pointing out that it is often impossible to identify insects without killing them because of their small size; they also emphasise the need to be able to identify species to monitor their population sizes. They echo Marshall (2006) when they state that amateur naturalists often do not need to kill insects, especially because 'identifying the general group of bees ... to which a bee belongs can be done *with* a little practice and *without* killing any bees.' Although Wilson and Carril are less cavalier about killing insects, they limit their concerns to species-level considerations:

Use your discretion. Nabbing one or two *Perdita* [a diverse genus of minute bees] for your collection is not, at this point, going to make the difference between a [bee species'] extinction and its survival. Nabbing a bumblebee, however, is not a good idea. In particular, when you collect a queen bumble bee (the giant ones that fly early in the year), you are in fact collecting an entire potential nest, as the queen is responsible for laying all the eggs that result in a colony of many individuals.

More generally, it is worth noting that animal welfare regulations do not require moral attention to insects as individuals: researchers do not, for instance, need

approval from an Institutional Animal Care and Use Committee (IACUC) to do research on most invertebrates, insects among them. By contrast, researchers *do* need such permission to work with vertebrates, and ethical codes concerning vertebrates tend to have the interests of individual animals in view, as seen in this part of an ornithological code:

Researchers involved in scientific collecting should assure that the manner of collection is *humane* and conforms to established *ethical standards of treatment* for animals in research. Collecting commonly entails trapping birds and employing a *humane method of euthanasia to bring about a rapid loss of consciousness and to minimise pain and suffering* prior to death. (Fair *et al.*, 2010, italics added)

A similar awareness of the welfare of individual animals exists among those who collect crustaceans:

Collecting crustaceans often means killing them. Although lamentable, we can conserve and protect only what we understand, and a thorough understanding in many cases means the detailed study of preserved specimens. Ironically, in order to save and preserve life on earth, we have to kill some of it. *Some methods and procedures are nevertheless more 'humane' than are others*, and some methods are destructive or wasteful. Often the most efficient methods of collecting are destructive ones, such as pounding apart pieces of coral, regrettably sometimes live. Collectors are urged to follow and recommend ethical procedures whenever possible. (Martin, 2016: 395)

Of course, there are excellent reasons to think that birds are conscious, and so it seems quite reasonable to insist on killing them humanely. However, the evidence for crustaceans being conscious is not obviously better than the evidence for insects being conscious. (See, for instance, the behavioral evidence that is surveyed in Gherardi, 2009 and the argument in Sneddon, 2015, much of which could also be said of insects.) If consciousness is a complex program to run, then it is significant that lobsters have roughly 1/10 the number of neurons as honeybees. It thus seems reasonable to extend the same kind of care to insects that have been extended to crustaceans.

Some moral risks of insect collection

Let us suppose, then, that entomologists ought to exercise some moral caution in their dealings with insects. Given as much, a number of moral concerns immediately arise.

First, insect sampling is often undertaken with little regard for their suffering. They are killed by numerous questionable means that would be regarded as inhuman for mammals, including intoxication and death by alcohol; using sticky traps (and then kerosene to detach

them); depriving them of food in emergence Malaise traps, which may only be checked infrequently; exposing them to ethyl acetate or potassium cyanide in a killing jar; or giving larger specimens a lethal injection. Of course, even if we assume that insects can feel pain, there remain legitimate questions about how painful these methods are, and we take no stance on that here. Our point is just that insect suffering appears not to be a consideration in the choice of these practices: they are designed to maximise the catch of desired species and to kill them in ways that preserve their usefulness for scientific purposes.

Second, most sampling techniques involve collecting and (intentionally or unintentionally) killing many individuals that are not members of the target taxon. There are, of course, methods that minimise this bycatch problem, such as using a pooter, yet they are inefficient in many contexts and essentially useless in others. So, the problem is not just that the *desired* insects are collected and killed in ways that give little consideration to the possibility of insect consciousness, but also that the methods of collection give little consideration to the importance of *minimising* the number of insects collected and killed. This does not fit neatly with the position of the British Entomological & Natural History Society, whose Conservation Working Group states that 'collecting should always be limited to the minimum necessary for the purpose intended' (Joint Committee for the Conservation of British Invertebrates, 2002).

Furthermore, the standard justifications for engaging in these practices are not entirely convincing. Drawing on several codes, the TEA (n/d) states two main 'benefits of collecting Lepidoptera and other insects' (with a third, collecting as 'a recreational activity,' being questioned by other codes cited earlier):

- 1 It is a means of introducing people, particularly children, to an awareness and study of an important part of their natural environment; and
- 2 It has an essential role in the elucidation of scientific information, both for its own sake and as a basis from which to develop rational means for protecting the environment and its resources.

With regard to the former, entomologists face the challenge of getting people to care about beings that they are killing (cf. Larson 2007). In general, they will face a public relations challenge if they wish to communicate the importance of insects yet kill them indiscriminately: a stance of respect for insect lives would be much more effective.

With regard to the latter, it is worth considering the paradox involved in killing for the purpose of conserving. Granted, scientific knowledge may be of value in its own right, and entomologists may want to learn about certain insect species not in order to save them, but to save other species that depend on them. Yet insofar as the goal is to conserve the very species whose members are being sampled, this behaviour is a bit strange. After all, there is an

important sense in which things cannot go better or worse for a species: a species is not a single entity whose welfare can be harmed. Instead, there is a live philosophical debate about whether a species is an abstraction, a construct that is useful for certain human goals (Stanford, 1995; Ereshefsky, 2017). That is not, of course, a reason not to care about species, but it is a reason to consider whether human interest in an abstraction can be justified when it comes at the expense of particular individuals (see also Powell, 2011).

Another point to keep in mind is that it is not just a matter of killing a few individuals to document and identify them because they cannot be identified when they are alive. That is, even individuals of fairly well-known species – unless they are large – will need to be killed forevermore to be identified. Entomologists need to continually revisit whether it is appropriate to institutionalise killing in their interactions with insects.

Mitigating moral risk

Of course, insect collection is an important practice for entomologists, and we are not arguing that it is never justified – or even that it is only rarely justified. However, given the possibility that insects can experience pain, it is worth considering whether ethical codes should include some guidelines that are motivated by the recognition of insects as individuals. These guidelines probably would not take the same form as the ones found in vertebrate or crustacean ethical codes, as the difference in confidence about consciousness is relevant to the constraints placed on research: higher confidence in consciousness – and so in the capacity to experience pain – justifies more significant restrictions; lower confidence in consciousness – and so more room for doubt about the capacity to experience pain – justifies correspondingly weaker restrictions.

This means that we are not recommending the ‘minimum ethic’ that Lockwood (1987, 1988) proposes, which requires that entomologists ‘refrain from actions which may be reasonably expected to kill or cause non-trivial pain to insects when avoiding these actions have no, or only trivial, cost to human welfare.’ Plainly, quite a bit of research on insects could be avoided without any likely cost to human welfare. Human welfare is at issue in the case of collecting mosquitoes that may bear malaria or insects that are destroying crops, but a great deal of it is motivated by more rarified interests, such as the attainment of knowledge itself. So, given the view that he defended in 1988 (which may not be his view now), entomologists are morally obligated to end most research on insects.

Though we do advocate a version of the precautionary principle – where caution is justified in causing harm in the context of uncertainty, we find Lockwood’s position too extreme. First, the benefits of research to human welfare are often uncertain, so too much inquiry would be threatened by this standard. Second, there remains

uncertainty that insects are conscious. Third, even if insects are conscious, it is not clear that an individual insect life is – or even many insect lives are – of sufficient value to warrant these kinds of restrictions on scientific research. After all, if information integration is the main purpose of consciousness, then it may well come in degrees. (This is actually entailed by certain theories of consciousness. Consider, for instance, Tononi’s (2008) Integrated Information Theory of consciousness, according to which consciousness just is an information-theoretic property. If this is right, then the more information in the system, and the greater its organisation, the greater the degree of consciousness.) If consciousness does come in degrees, then given the behavioural evidence, it is highly unlikely that insects are conscious to nearly the degree that human beings are. It would be more difficult to assess whether particular taxa (at particular lifestages) would have roughly the same degree of consciousness as some of the least sophisticated vertebrate animals (e.g., sardines). Still, if moral importance tracks degree of consciousness, then it is unlikely that insects would deserve the same kind of moral protection extended to, say, small mammals, such as the mice regularly used in research. Fourth, we note that the simpler an organism’s cognitive capacities, the less psychologically connected it is to its own future. And if it is plausible that death is worse for beings with more complex psychologies – ones that permit them to form a conception of themselves as extended in time – then it follows that death is less bad for simpler organisms.

We recommend a more modest position. Recall the 3Rs, which recommend replacing animals with non-living models, reducing the number of animals used, and refining animal care and use practices. The underlying thought behind these guidelines is that the interests of animals are morally important, and while they may matter significantly less than human interests – after all, society would never sanction non-therapeutic research on human beings – they matter enough to justify expenditures for their sake. It can be expensive to find new models, to slow down research, and to improve the conditions under which animals are kept and used, yet these are considered acceptable costs given the moral significance of animal interests.

If that is what morality requires in the case of animals that researchers generally agree are sentient, then it seems reasonable to ask for a more moderate version of this approach where there is less certainty. That is, a modified version of the 3R framework will allow that non-therapeutic considerations can justify harm to insects, and the bar for adequate justification will be lower than it is for research on animals where the evidence of consciousness is much clearer. However, it will also acknowledge that there is a certain kind of moral risk associated with those harms, and so it urges entomologists to minimise that risk relative to their aims. Accordingly, we propose that entomologists should aspire to study insects without killing them, to reduce the number of insects they kill, and to

refine their methods so that when they do capture or kill insects, they do so in ways that are sensitive to their subjects' well-being. (This last goal might require the kind of work that Dawkins, 1990 and others have done with vertebrates, that is, conducting research on the revealed preferences of insects.) Again, the moral pressure to do these things is weakened by a lack of confidence about insect consciousness. But relatively weak pressure is still some pressure, and worthy of some attention.

Of course, others will object that there is not enough pressure. Shouldn't a code of ethics be more demanding? If there is any real possibility that insects are conscious, shouldn't every precaution be taken? Obviously, that is not our view. As we said at the outset, there are two sets of concerns that are relevant to an ethics code. First, there are baseline expectations, which should be driven by widely accepted values both within and beyond the field. Second, there are concerns about mitigating moral risk. This is where we consider the possibility of insect consciousness: the evidence for that hypothesis is not sufficient to let it drive baseline expectations among entomologists. But it is sufficient to make it a live possibility, which means that ethical codes – and thus practices – ought to be adjusted accordingly.

The example of malaise traps

We next consider a case study that will help to clarify what it might mean to employ the 3Rs in the context of insect biodiversity sampling. Consider Malaise traps, which are often the only practical way to collect many insects, yet have enormous bycatch rates. If the focus is species preservation, then there will be many contexts in which such rates are of little concern: the kinds of insects being caught are likely to be sufficiently plentiful. That said, given increasing concerns about insect population declines this should not just be assumed: insect collection is often justified with the claim that entomologists know so little about insect diversity that collection is necessary to improve knowledge; however, the converse is that entomologists often do not know with any certainty the effect of their sampling on insect populations. Regardless of these considerations, bycatch is of significant concern when individuals are in view. If the goal is to study bees, and a sampling regime catches five times as many moths as bees, then even if the reasons for engaging in the research are sufficient to justify killing those bees, it does not immediately follow that they are sufficient to justify killing all the moths as well.

When we consider any particular researcher's deliberations about whether to use a Malaise trap, it may appear that the 3Rs would simply discourage their use. If bees are themselves the object of inquiry, however, then replacement is unlikely. This leaves the options of reduction and refinement. There are ways to reduce the amount of bycatch in pan traps, such as by colouring them to be more attractive to a target species, which reduces the

amount of time that the trap needs to be present. However, while Malaise traps can be made larger or smaller, and they can be placed in different locations, strategies that reduce the number of non-target species that are caught also reduce the number of target species that are caught. This can mean needing to have the trap in place for a longer period of time, which of course increases the number of non-target deaths. And although there are different killing agents available, it is not obvious that any of them put insects at less risk of experiencing pain – if, in fact, they can experience pain.

All that said, recall that uncertainty lowers the standard of justification. There may be good reasons to pursue this particular research project. Yet, based on the 3Rs, researchers should seek to reduce bycatch and reduce potential suffering as much as possible. But how?

Malaise traps may be used for several reasons, including the detection of new species and the documentation of trends in insect biodiversity. In both instances, entomologists can first consider whether such sampling might be replaced – or even eliminated. Arguably, in many systems researchers already know enough about insect biodiversity and its decline to make a case for insect conservation (Hallmann *et al.*, 2017), and entomologists would have a greater impact by focusing their energies on working in the 'popular' and political realm (e.g., to argue for bans on neonicotinoids) rather than in fulfilling their scientific curiosity. Entomologists might also reduce sampling effort in a given location to lower rates of bycatch, even if this means a lower likelihood of detecting new species; at a certain point, the wish to document existing diversity – before it disappears – must be weighed against the need to do everything possible to reduce the potential that it is, in fact, lost for ever.

Entomologists might also ask whether the use of Malaise traps – and the unintentional killing of many non-target species – generates other obligations on the research community generally. For instance, if researchers choose to use Malaise traps, it may be important for them to develop a registry to coordinate a global sampling regime and to create opportunities for other researchers to study whatever the trap collects. Since the bycatch for one researcher can be the target species for another, as Spears and Ramirez (2015) observe, this can lead to a net reduction of individual insects collected. Granted, this is not a trivial burden to place on researchers: it takes work to build a registry of this kind, and it delays some research projects as it would be necessary to coordinate with more people. Nonetheless, this might be exactly what the 3Rs require. Moreover, the fact that some entomologists already do this is evidence that it can be done in a way that is not overly burdensome.

Conclusion

Whenever researchers are considering capturing or doing research on insects, they ought to ask some questions with all sincerity. Is it possible to do this research using computer

simulations or existing insect collections? Could it be done with fewer insects? And if insects are going to be captured and/or harmed, how burdensome would it be to reduce the risk of suffering? These questions should be asked alongside a consideration of the human interests served by the relevant endeavour, as well as the promise of the research program itself. If the human interest is a weighty one, then that counts in favour of a more permissive approach to insect use. If, however, the interest is more like a mild preference, then it may not rise to the level of a concern that would seem reasonable to others. Likewise, if researchers have greater confidence that the methods will deliver the intended answers and applications, then it is easier to justify collection and research. And if, on the other hand, a method is one that is unlikely to produce much in the way of knowledge, or technological progress, or what have you – based on a sincere and evenhanded assessment of the odds – then it is correspondingly harder to justify.

We acknowledge that the use of the 3Rs requires judgment. Moreover, the field will need to grapple, in its own way, with the thorny questions of interpretation with which other fields have been grappling for a long time (e.g., potential ambiguities in the various Rs, see Olsson *et al.*, 2012). So there is no simple decision procedure that we can recommend, and the field will need to determine – largely on its own, but partially in conversation with interested outsiders – the sorts of expectations that are reasonable. Our aim here is simply to encourage entomologists to see these conversations as relevant to their field, to enter them, and to debate appropriate actions.

Acknowledgements

We appreciate comments on the manuscript from Clare Palmer, the associate editor, and anonymous reviewers.

Conflict of interest

The authors have no conflicts of interest to report.

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Accepted 4 February 2019

First published online 25 February 2019

Editor: Raphael Didham

Associate editor: Francis Gilbert